MITOCHONDRIAL & CELLULAR ENERGY

# Glutathione — the full review

ENDOGENOUS TRIPEPTIDE ANTIOXIDANT

*This is the deeper look behind the short version — what glutathione does in the body, why the oral form is a poor route, what the skin-lightening and Parkinson's studies actually measured, the safety record in detail, who it is not for in full, and what your consultation covers.*

## What it is

Glutathione is a tiny protein of three amino acids that every cell in the body manufactures for itself. It is the body's main internal antioxidant and its principal tool for neutralizing reactive by-products of metabolism, including in the liver.

Levels fall with illness and with age, which is part of why it is used for skin tone, oxidative stress, and general recovery. Swallowed glutathione is broken down before it can be absorbed intact, which is the reason it is given by injection, IV, or nebulizer instead of as an oral supplement.

## What people use it for

| USED FOR | EVIDENCE | WHAT THAT MEANS |
|---|---|---|
| **General antioxidant and oxidative stress** | **Limited human data** | Small human studies measured blood markers rather than outcomes. |
| **Skin tone and brightening** | **Limited human data** | The most-marketed use; the trials behind it are small and mixed. |
| **Chemotherapy side effects and neuropathy** | **Limited human data** | Studied as an add-on in oncology, with results that need an oncologist's input. |

## How it works

As the body's primary intracellular antioxidant, glutathione works by donating electrons to neutralize reactive oxygen species and other reactive by-products of normal metabolism, and it plays a central role in liver detoxification pathways. Because oral glutathione is largely broken down by digestion before it can be absorbed intact, routes that bypass the digestive system — injection, intravenous infusion, or nebulized inhalation — are used instead.

The proposed skin-brightening effect is thought to work through glutathione's influence on melanin production, shifting pigment production toward a lighter form of melanin, though the underlying biochemistry is better established in laboratory settings than confirmed in controlled human skin studies.

## What the research actually looked at

Small human studies of glutathione have generally measured blood markers of oxidative stress rather than direct clinical outcomes. Skin tone and brightening is the most heavily marketed use, but the trials behind it are small and their results are mixed.

Glutathione has also been studied as an add-on therapy for chemotherapy side effects and chemotherapy-induced neuropathy in oncology settings, where results require interpretation

alongside a patient's oncologist. Across all these uses, the human evidence base is limited in size and consistency relative to how widely the compound is marketed.

### Common misunderstandings

- **“Oral glutathione supplements raise your levels the way an infusion does.”** Swallowed glutathione is largely broken down during digestion before it can be absorbed intact, which is why injectable, IV, and nebulized forms are used instead.
- **“It's a proven skin-lightening treatment.”** The trials behind the skin-brightening use are small and their results are mixed, not a settled, consistent finding.
- **“It's approved for detoxification or wellness use.”** No form of glutathione is FDA-approved for skin brightening, detoxification, or general wellness use.
- **“It's risk-free because it's a substance the body already makes.”** Naturally occurring does not mean risk-free by injection; reactions, and contamination in compounded product, are real considerations.

### Who it is not for

- Anyone with asthma or reactive airways, where an inhaled or nebulised form is proposed.
- Anyone with a history of anaphylaxis or severe drug allergy.
- Anyone with liver disease or abnormal liver tests.
- Pregnant or breastfeeding patients, and anyone on chemotherapy without involving their oncologist.

**Competitive athletes:** Glutathione itself is not prohibited by WADA. Athletes in tested sport should be aware that WADA's M2 category restricts large-volume intravenous infusions (generally above 100 mL within a 12-hour period) except under specific permitted circumstances, which can affect how an infusion is administered rather than whether glutathione itself is allowed.

### Who makes this, and under what rules

**Everything dispensed here is prepared by a licensed 503A compounding pharmacy** — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

**A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.**

## Where this stands

**Not approved for the uses it is marketed for.** Injectable and IV glutathione is widely offered for skin brightening, detoxification and wellness. None is an FDA-approved use.

**Human outcome evidence is limited.** Regulators in several countries have warned about injectable skin-lightening products specifically.

**Well tolerated is not the same as effective. Both need to be true.**

### Clinical detail

A tripeptide (glutamate-cysteine-glycine) synthesized endogenously in every human cell; the body's principal intracellular antioxidant.

Levels decline with illness and age.

Oral bioavailability is poor; clinical and wellness use relies on injection, IV infusion, or nebulization.

Studied as an oncology add-on for chemotherapy-induced neuropathy and other side effects.

Contamination and sterility issues have been a recurring problem in compounded glutathione products.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

Glutathione is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.